Math Kangaroo Practice - Answer Key

For practice_2002_2003.pdf (Selected problems from 2002 (#4, #7, #9) and 2003 (#9, #12, #14))

Answer key & explanations

2002 #4: D The pieces are: 1 small square, 2 dominoes, an L of 3, a T of 4 and an L of 4. There is no U-shaped piece.

1.1: B Pieces: two L-shapes of 4, a straight 3, two dominoes, one small square. No T-shape.

1.2: C Pieces: four L-shapes of 4, one T, one L of 3, one domino. There is no zig-zag (S) piece.

1.3: E Pieces: three L-shapes of 4, one T, two L-shapes of 3, one straight 3. No 2x2 square.

2002 #7: C Wherever there are two arches, they sit right next to each other. There is never a U-shaped dip between two arches.

2.1: A In the skyline there is a flat part between the wall and the pointed roof. Piece A has the roof sitting straight on the wall.

2.2: E The castle wall has TWO notches at the top. Piece E shows a whole tower top with only one notch.

2.3: D After each arch the line goes flat and then UP. Piece D goes down after the arch.

2002 #9: E 6 oranges + 1 melon = 2 melons. Take one melon off each side: 6 oranges = 1 melon.

3.1: C 8 plums + 1 lemon = 3 lemons, so 8 plums = 2 lemons, so 1 lemon = 4 plums.

3.2: D 10 cherries + 2 apples = 4 apples, so 10 cherries = 2 apples, so 1 apple = 5 cherries.

3.3: D Take 3 oranges off each side: 1 melon = 6 oranges, so 2 melons = 12 oranges.

2003 #9: A Brick is 5 x 4 x 4. Remove the outer layer: inside is 3 x 2 x 2 = 12 blue cubes.

4.1: B Inside is (4-2) x (4-2) x (3-2) = 2 x 2 x 1 = 4.

4.2: C Inside is (6-2) x (3-2) x (4-2) = 4 x 1 x 2 = 8.

4.3: D Total = 5 x 3 x 3 = 45. Inside (blue) = 3 x 1 x 1 = 3. Red = 45 - 3 = 42.

2003 #12: A Stand in front of the door wall: the dot wall is on your LEFT. So from the back (no door), the dot is on the right of the plain end wall - house A.

5.1: C The door is on the end wall that has the roof attached. Facing that door, the dot wall is on your left.

5.2: D Facing the door wall, the window wall is on your right and the plain wall on your left. Only D shows that.

5.3: B Facing the dot end wall: the door wall is on your right, the window wall on your left. B shows the dot wall with the door wall to its right.

2003 #14: E Each side bar is 6. The slanted part is a parallelogram 1 wide and 6 tall = 6 (cut-off triangles fit together). 6+6+6 = 18.

6.1: C Top bar 5 + bottom bar 5 + slanted part (2 wide, 5 tall) 10 = 20.

6.2: C Without the notch the shape is a 2 x 5 rectangle (10) plus two side triangles (each half of 2 x 5 = 5), so 20. The white triangle notch at the top is half of 2 x 3 = 3. 20 - 3 = 17.

6.3: C Two side bars: 5 + 5 = 10. The middle V is two slanted strips, each 3 across and 2 tall = 6. 10 + 6 + 6 = 22.